



PHASE 8

Testing and Evolution

Test strategy, limitations, roadmap and release readiness

SMARTPOCKET AI • SMART BUDGET AND RECOMMENDATION ASSISTANT

1. Testing and Evolution

1.1 Test Matrix

Area	Essential tests
Authentication	Duplicate account, invalid password, expired token, protected route without token.
Validation	Zero or negative budget, missing fields, invalid guest count, unsupported image payload.
Home planner	Different room selections, zero quantities where allowed, budget boundary checks.
Party planner	Guest limits, optional services, venue response shape, total allocation.
Jewelry planner	Text-only request, image request, invalid image, outfit analysis shape.
Normalization	Over-budget AI result, negative prices, missing links, alternate keys, rounding excess.
History	Create, list, newest-first order, reuse, owned delete, cross-user access denial.
Responsive UI	Navigation, forms, cards, tables and errors on common viewport sizes.

1.2 Current Limitations

- Prices are generated estimates unless a verified live product API is integrated.
- Search links may lead to changing marketplace results.
- The service depends on network access and AI-provider availability.
- SQLite is appropriate for a local or small deployment but should be reviewed for higher concurrency.
- Browser token storage requires strong protection against cross-site scripting.

1.3 Recommended Evolution

- Integrate approved live product and price feeds.
- Add explicit refresh-token and revocation workflows.
- Move production persistence to a managed relational database.
- Add rate limiting, audit logs, tracing, and structured metrics.
- Provide exportable plan summaries and visual budget charts.
- Extend planner modules through reusable schemas and route templates.
- Add accessibility validation and multilingual interface support.

1.4 Release Readiness Checklist

- All secrets are externalized and rotated.
- Automated tests cover the budget guarantee and ownership checks.



- CORS allows only deployed frontend origins.
- Database backup and migration procedures are documented.
- Error messages are safe, actionable, and free of stack traces.
- Mobile and desktop flows have been manually verified.
- AI model, timeout, quota, and fallback behavior are documented.

Final implementation position

SmartPocket AI is strongest when generative recommendations remain advisory and every value used by the product is validated by deterministic application logic. This design keeps the experience flexible while preserving budget, security, and data-ownership rules.